

The linear-logic tensors: $\ltimes$ and $\rtimes$ as Dialectica products

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1.1 Setup: is there anything outside the Day family?

Recall the standing conventions. **Cont** is the category of containers, equivalent to the polynomial functors **Poly**; a container $p = (S_p, p[-])$ has a set S_p of shapes and, at each shape s , a set $p[s]$ of directions (positions), with extension $\llbracket p \rrbracket X = \sum_{s \in S_p} X^{p[s]}$. We write y^A for the representable with one shape and directions A , and $y := y^1$ for the unit container. Four monoidal structures are the standard census: the coproduct $+$, the categorical product $\times$, the Dirichlet tensor $\otimes$, and the sequential (substitution) operator $\triangleleft$.

The classification theorem (Theorem A) accounts for the *Day-convolutional* part of this census. Call a monoidal structure $(\odot, J)$ on **Cont** *convolutional* when $\odot$ preserves coproducts in each variable and carries representables to representables; equivalently $(p \odot q)[(s, t)] = p[s] \star q[t]$ for a monoidal structure $(\star, I)$ on **Set**, of which $\odot$ is the Day convolution. Theorem A says the Day construction $(\star, I) \mapsto (\odot_\star, y^I)$ is an *equivalence* between monoidal structures on **Set** and convolutional monoidal structures on **Cont**; $\times$ and $\otimes$ are the images of $(+, \emptyset)$ and $(\times, 1)$, while $+$ and $\triangleleft$ sit outside for elementary reasons.

An equivalence onto the *convolutional* structures is, honestly read, a statement about a sub-family. It leaves a question open at the level of **Cont** itself:

*Are there monoidal structures on **Cont** outside the Day family altogether — tensors whose direction at (s, t) is **not** a pointwise function $F(p[s], q[t])$ of the two fibres?*

The answer is yes, and the examples are not exotic: they were written down by Dorta–Jarvis–Niu [2] and left *uninterpreted*. This section identifies them as the Dialectica tensors of Gödel’s functional interpretation, answering DJN’s own stated open question.

1.2 The two DJN tensors in container coordinates

Dorta–Jarvis–Niu study monoidal structures on generalised polynomial categories $\Sigma\Pi\mathcal{C}$; specialising to $\mathcal{C} = \mathbf{1}$ gives $\Sigma\Pi\mathbf{1} \simeq \mathbf{Poly} \simeq \mathbf{Cont}$ [2, Prop. 2.9]. Beyond the Day convolution $\otimes$ and the composition product $\triangleleft$, their §6 exhibits *at least two further* monoidal structures with the same unit y , which we write $\ltimes$ and $\rtimes$.

Definition 1.1 ($\ltimes$ and $\rtimes$, in container coordinates). Both tensors have shape set $S_p \times S_q$. Their directions are

$$(p \ltimes q)[(s, t)] = p[s]^{S_q} \times q[t]^{S_p}, \quad (p \rtimes q)[(s, t)] = p[s]^{S_q} \times q[t],$$

with unit y on both sides. Here X^S denotes the function set $S \rightarrow X$, and the exponent S is the *shape set of the other factor*. [Cited: DJN [2, §6], in container coordinates]

The tensor $\ltimes$ is *symmetric*: each factor’s direction set is exponentiated by the other factor’s shape set. The tensor $\ltimes$ is *triangular*: only the left factor is exponentiated, so it will turn out to be associative but not symmetric (Lemma 1.2). These are two of the monoidal products DJN define; they explicitly leave their meaning open, writing that “we would like to know if there are interpretations or applications for these monoidal products” [2].

The single feature that governs everything below is the exponential $(-)^{S_q}$: a direction set exponentiated by a *global* shape set, rather than the pointwise $p[s] \star q[t]$ of a Day tensor. That is what pushes $\ltimes/\ltimes$ outside the Day family (§1.5), and it is the Dialectica hallmark (§1.4).

Lemma 1.2 (Unit laws and closed forms). $p \ltimes y \cong p \cong y \ltimes p$ and $p \ltimes y \cong p \cong y \ltimes p$. The k -fold iterates have closed forms

$$(p_1 \ltimes \cdots \ltimes p_k)[(s_1, \dots, s_k)] \cong \prod_{f=1}^k p_f[s_f]^{(\prod_{g \neq f} S_{p_g})},$$

$$(p_1 \ltimes \cdots \ltimes p_k)[(s_1, \dots, s_k)] \cong \prod_{f=1}^k p_f[s_f]^{(\prod_{g > f} S_{p_g})}.$$

Consequently $\ltimes$ is associative and symmetric, while $\ltimes$ is associative but not symmetric. [MacBeth (computed; exponential laws)]

Proof. For the unit, $S_y = 1$ and $y[*] = 1$, so $(p \ltimes y)[(s, *)] = p[s]^1 \times 1^{S_p} = p[s]$, and symmetrically; the $\ltimes$ case is the same computation. The closed forms follow by induction from Definition 1.1 using $(X \times Y)^S \cong X^S \times Y^S$ and $X^{S \times T} \cong (X^S)^T$. In the $\ltimes$ form each factor p_f is exponentiated by the product of *all* other shape sets, an expression invariant under reassociation and under permutation of the factors; hence $\ltimes$ is associative and symmetric. In the $\ltimes$ form each factor is exponentiated only by the shape sets *to its right*, an expression unambiguous under reassociation (the product over $g > f$ does not depend on bracketing) but sensitive to the factor order; hence $\ltimes$ is associative but not symmetric. $\square$

The triangular exponent pattern is the precise sense in which $\ltimes$ is *directed*: reading the factors left to right, factor f is challenged by the choices of all *later* factors, but not the earlier ones — a strict, one-directional dependency, of which $\ltimes$ is the symmetric closure.

1.3 The homogeneous fragment is de Paiva’s Dialectica category

Definition 1.3 (Homogeneous container). A container is *homogeneous* of arity (I, A) if it has the form $p = \sum_{i \in I} y^A$: shape set I , with *every* direction set equal to the same A . [Cited: DJN [2, §2]]

DJN prove that the homogeneous objects form a full subcategory equivalent to de Paiva’s original Dialectica category [2, Prop. 2.13]:

$$\text{Hmg}(2) \simeq \mathbf{Dial}(\mathbf{Set}).$$

Under this dictionary a homogeneous object corresponds to a de Paiva triple (U, X, α) , where U is the set of *positions* (the shape set I), X is the set of *challenges* (the common direction set A), and α is a set-valued (relational) predicate on $U \times X$. A morphism carries a forward map $f: U \rightarrow V$ together with a backward map $F: U \times Y \rightarrow X$ — exactly the two components of a container morphism between homogeneous objects. So *positions are U and directions are X* .

Definition 1.4 (de Paiva’s Dialectica tensor). On $\mathbf{Dial}(\mathbf{Set})$ the multiplicative (linear-logic) tensor is

$$(U, X, \alpha) \otimes_{\mathbf{Dial}} (V, Y, \beta) = (U \times V, X^V \times Y^U, \alpha \otimes \beta),$$

where, for $f: V \rightarrow X$ and $g: U \rightarrow Y$, the predicate is $(\alpha \otimes \beta)((u, v), (f, g)) = \alpha(u, fv) \wedge \beta(v, gu)$. [Cited: de Paiva [1]]

The witness object $X^V \times Y^U$ is the defining feature of the Gödel–de Paiva functional interpretation: to answer a challenge against a conjunction $A \otimes B$ one must supply, for each of the opponent’s moves in one component, a response in the *other* — hence the mutual function spaces.

1.4 Main identification: $\ltimes$ is de Paiva’s tensor, extended

Proposition 1.5 ($\ltimes$ restricts to the Dialectica tensor). *On the homogeneous fragment $\text{Hmg}(2) \simeq \text{Dial}(\mathbf{Set})$, the tensor $\ltimes$ restricts to de Paiva’s Dialectica tensor $\otimes_{\text{Dial}}$. Concretely, for homogeneous $p = \sum_I y^A$ and $q = \sum_J y^B$,*

$$p \ltimes q = \sum_{I \times J} y^{A^J \times B^I},$$

whose shape set $I \times J = U \times V$ and direction set $A^J \times B^I = X^V \times Y^U$ match Definition 1.4 exactly. Hence $\ltimes$ is de Paiva’s Dialectica tensor extended off $\text{Hmg}(2)$ to all of $\mathbf{Poly} \simeq \mathbf{Cont}$. [MacBeth (an identification; matches definitions)]

Proof. Take p and q homogeneous with $S_p = I$, $p[i] = A$ for all i , and $S_q = J$, $q[j] = B$ for all j . Definition 1.1 gives shape set $I \times J$ and, at every shape (i, j) ,

$$(p \ltimes q)[(i, j)] = p[i]^{S_q} \times q[j]^{S_p} = A^J \times B^I.$$

This direction set is independent of (i, j) , so $p \ltimes q$ is again homogeneous, of arity $(I \times J, A^J \times B^I)$, i.e. $p \ltimes q = \sum_{I \times J} y^{A^J \times B^I}$. Under the dictionary of §1.3 — $U = I$, $V = J$, $X = A$, $Y = B$ — the shape set is $U \times V$ and the direction set is $X^V \times Y^U$, the forward and backward objects of $\otimes_{\text{Dial}}$ in Definition 1.4. The predicate matches by the same substitution: a direction $(f, g) \in A^J \times B^I$ is precisely a pair $f: V \rightarrow X$, $g: U \rightarrow Y$, and DJN’s fibrewise predicate composition realises $\alpha \otimes \beta$. So on $\text{Hmg}(2)$ the two tensors agree, and $\ltimes$ is the extension of $\otimes_{\text{Dial}}$ to all containers. $\square$

For $\ltimes$ the corresponding restriction gives, on homogeneous inputs, direction $A^J \times B$: the left challenge is functionalised over V but the right challenge is presented bare. By Lemma 1.2 the k -fold $\ltimes$ exponentiates each factor’s challenge only over the positions of the factors to its *right*. So $\ltimes$ is the *directed* Dialectica tensor: where $\ltimes$ is the conjunction in which both players respond adaptively to the other’s move, $\ltimes$ is the conjunction whose dependency flows one way only — one problem answered adaptively in response to the other, which is played blind. Contrast the two closed forms of Lemma 1.2: $\ltimes$ exponentiates over the product of *all* other shape sets and is symmetric; $\ltimes$ exponentiates only rightward and is directed.

For orientation, here is how the two Dialectica conjunctions split between the Cont tensors. On $\text{Hmg}(2)$ the Day tensor $\otimes$ restricts to direction $A \times B = X \times Y$ — the naive Gödel conjunction $A \wedge B$, whose challenge is the plain product of challenges — whereas $\ltimes$ restricts to the linear tensor $X^V \times Y^U$. The two structures cleanly separate the non-linear ($\otimes$) and multiplicative-linear ($\ltimes$) conjunctions of the Dialectica interpretation.

1.5 Structural consequences

Proposition 1.6 (Non-convolutional). *Neither $\ltimes$ nor $\ltimes$ is convolutional: there is no functor $F: \mathbf{Set} \times \mathbf{Set} \rightarrow \mathbf{Set}$ with $(p \ltimes q)[(s, t)] = F(p[s], q[t])$. In particular they are not Day convolutions of any monoidal structure on $\mathbf{Set}$, so Theorem A does not reach them. [MacBeth (computed witness)]*

Proof. Let $p = y^2$ (one shape, direction set of size 2) and, for each n , let q_n be homogeneous with $|S_{q_n}| = n$ and every direction set of size 2. Then $(p \ltimes q_n)[(*, t)] = 2^n \times 2^1 = 2^{n+1}$. The two fibres $p[*] = 2$ and $q_n[t] = 2$ are fixed independently of n , yet the resulting direction

set grows with $n = |S_{q_n}|$. A convolutional formula $F(2, 2)$ would be constant in n ; no such F exists. The same inputs defeat $\bowtie$. Since every Day tensor is convolutional (with $F = \star$), neither tensor is a Day convolution, and Theorem A — an equivalence onto the *convolutional* structures — does not apply to them. $\square$

This is the honest boundary of the classification. The Day family — the four canonical tensors together with the wider family indexed by monoidal structures on **Set** — does *not* exhaust the monoidal structures on **Cont**. The structures it omits are the linear-logic tensors $\ltimes$ and $\rtimes$, and the reason it omits them is precisely that their direction is a non-pointwise, shape-set-indexed exponential rather than a fibrewise formula.

Proposition 1.7 (Non-cocontinuous). $(-) \ltimes q$ does not preserve coproducts: $\ltimes$ does not distribute over $+$. The same holds for $\rtimes$. Conjecturally, neither tensor is closed. [MacBeth (computed); closure Open]

Proof. The shape set of $p + p'$ is $S_p + S_{p'}$, so $((p + p') \ltimes q)[(s, t)]$ has the exponent $q[t]^{S_p + S_{p'}} \cong q[t]^{S_p} \times q[t]^{S_{p'}}$, whereas the p -summand of $(p \ltimes q) + (p' \ltimes q)$ has exponent $q[t]^{S_p}$. These differ (for a concrete instance the direction cardinality profiles are $[8, 8, 32, 32, 32, 32]$ against $[8, 8, 16, 16, 16, 16]$), so there is no natural distributive isomorphism. The obstruction is exactly the exponent's dependence on the *entire* shape set S_p , which a coproduct enlarges. A tensor whose left action fails to preserve coproducts cannot have a right adjoint in that variable, so $(-) \ltimes q$ is not a left adjoint and $\ltimes$ is not left-closed; we record the closure of $\rtimes$ as conjecturally negative for the same reason. $\square$

These are, to our knowledge, the first non-closed monoidal structures in the **Cont** story, in sharp contrast with the census: $\times$ is cartesian closed, $\otimes$ is closed, and $\triangleleft$ has a right co-closure. Because $\ltimes/\rtimes$ are non-convolutional, this non-closure says nothing about whether the *convolutional* tensors are closed; the two questions are disjoint.

1.6 Related work

We are careful about what is new here, because the theme is well populated. What this section contributes is an *identification* — DJN's uninterpreted $\ltimes$ is de Paiva's Dialectica tensor, extended off the homogeneous slice — together with the closed forms of Lemma 1.2 that expose the directedness of $\rtimes$, and the taxonomy placement of §1.5. It is emphatically *not* the first connection between Dialectica and polynomial functors; that relationship is the animating theme of DJN itself.

The following distinctions matter.

Lucatelli Nunes–Vákár [3] give a Dialectica-formula monoidal *closed* structure on Grothendieck constructions, which instantiates to containers. But their tensor is the *fibred product*, and the Dialectica twist lives in their internal hom $\multimap$, not in the tensor. They neither produce $\ltimes$ nor address DJN's open question; their construction is orthogonal to the identification made here.

Capucci–Gavranović–Malik–Rios–Weinberger [4] unify lenses, optics and Dialectica fibrationally — at the level of the *category*, as a common fibrational pattern, not as a monoidal tensor on **Cont**. Again this is a different object from $\ltimes$.

Finally, one must pre-empt the objection that DJN's own abstract already “extends dialectica categories”. It does — but their extending tensor (their §3) is the *non-twisting* parallel product, with direction $A \times B$; that is the Gödel conjunction, not the multiplicative one. The genuinely twisting multiplicative tensor, with the mutual function spaces $X^V \times Y^U$, is $\rtimes$, and $\ltimes$ is exactly what DJN left uninterpreted. It is that gap this section fills, framed as an answer to DJN's own stated open question.

Remark 1.8 (Grade). The identification of Proposition 1.5 is a matching of definitions and is *proved* at the level of the formulas, not conjectural. Its *novelty* rests on the observation that DJN, who know **Dial(Set)** $\simeq$ Hmg(2) intimately, nonetheless posed the interpretation of $\ltimes/\rtimes$ as open; a full novelty check against the recent Dialectica–polynomial literature (de Paiva, Trotta, Spivak, Hedges, 2023–2026) is owed before publication. We grade the mathematics *computed/an identification* and flag the novelty as pending that check.

References

- [1] V. de Paiva, *The Dialectica Categories*, in *Categories in Computer Science and Logic* (Boulder, 1987), Contemp. Math. **92**, Amer. Math. Soc., 1989, pp. 47–62; see also *The Dialectica Categories*, Ph.D. thesis / Tech. Report 213, Univ. of Cambridge Computer Laboratory, 1991.
- [2] G. Dorta, P. Jarvis, and N. Niu, *Monoidal Structures on Generalized Polynomial Categories*, arXiv:2305.05655; EPTCS **397** (2023).
- [3] F. Lucatelli Nunes and M. Vákár, *Dialectica Logical Principles and the Dialectica Construction on Grothendieck Fibrations*, arXiv:2405.07724 (2024).
- [4] M. Capucci, B. Gavranović, J. Malik, F. Rios, and P. Weinberger, *A Fibrational Account of Lenses, Optics, and Dialectica*, Proc. Math. Found. Program. Semantics (MFPS) **40** (2024).
- [5] N. Niu and D. I. Spivak, *Polynomial Functors: A Mathematical Theory of Interaction*, arXiv:2312.00990 (2023).